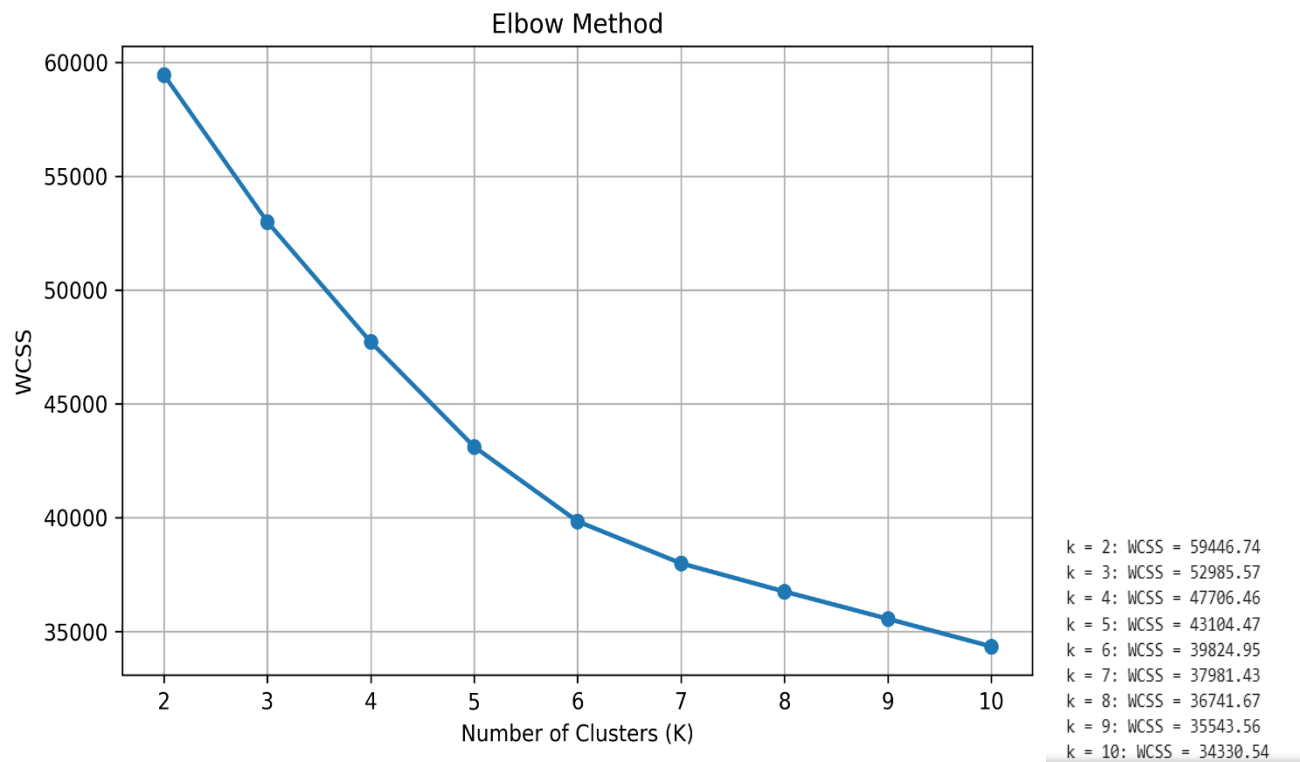
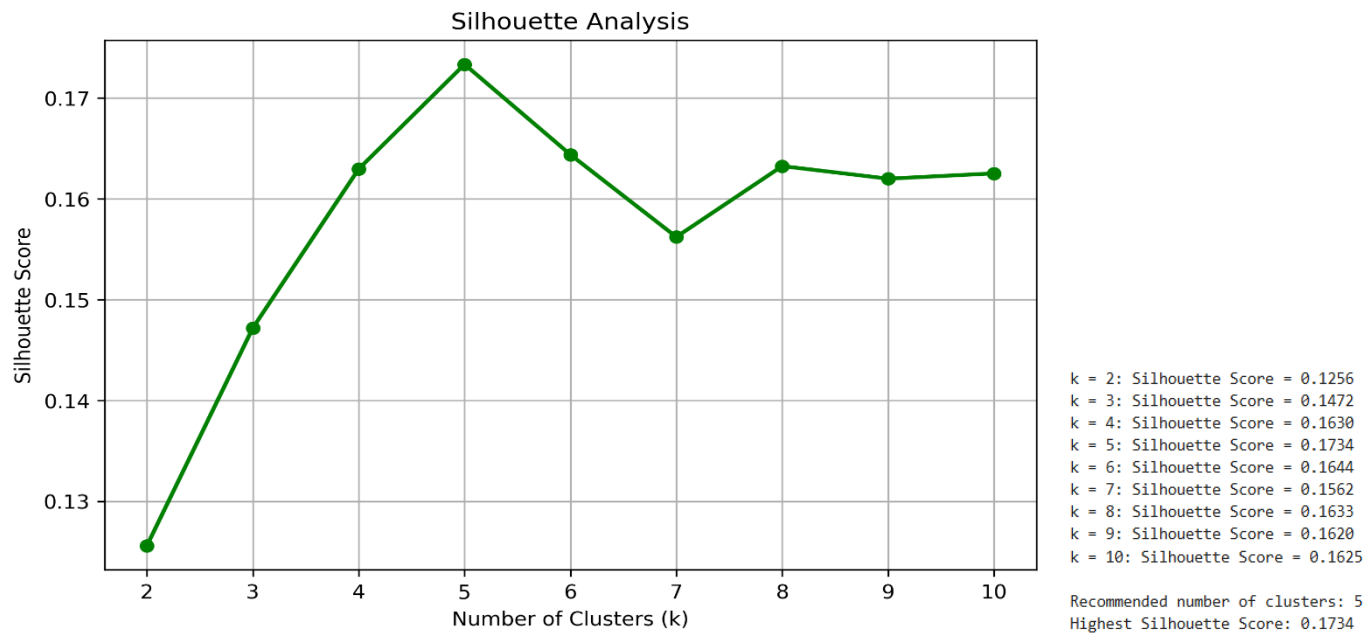


Optimal number of Cluster

The determination of the optimal number of clusters for the Customer Churn Analysis was guided by a dual-method evaluation incorporating both the Elbow Method and Silhouette Analysis. These techniques provide complementary perspectives on cluster validity, enabling a more robust and defensible selection of k .



The Elbow Method plot demonstrates a pronounced reduction in Within-Cluster Sum of Squares (WCSS) for lower values of k , followed by a gradual flattening of the curve. The point at which this transition occurs—specifically around $k = 3$ —indicates that additional clusters beyond this threshold contribute only marginal improvements to intra-cluster compactness. This inflection point is widely recognised as a reliable heuristic for identifying an efficient and parsimonious clustering structure.



The Silhouette Analysis further substantiates this conclusion. Although silhouette scores exhibit variation across the tested range of k , the score associated with $k = 3$ reflects a balanced compromise between cluster cohesion and separation. While marginally higher silhouette values may appear at other points, these do not correspond with meaningful structural gains when considered alongside the diminishing returns observed in the Elbow Method. Thus, the convergence of evidence across both diagnostic tools supports the selection of **three clusters** as the most analytically appropriate configuration.

Collectively, these visual and statistical assessments indicate that a three-cluster solution provides a coherent, interpretable, and methodologically justified segmentation of customer churn behaviour.